

ASSIGNMENT 2

1. To capture and analyze ICMP packets (Ping request and reply)

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harsh — zsh — 100x35

[(base) harsh@Harshs-MacBook-Air ~ % ping google.com
PING google.com (142.250.182.238): 56 data bytes
64 bytes from 142.250.182.238: icmp_seq=0 ttl=118 time=16.323 ms
64 bytes from 142.250.182.238: icmp_seq=1 ttl=118 time=59.844 ms
64 bytes from 142.250.182.238: icmp_seq=2 ttl=118 time=15.463 ms
64 bytes from 142.250.182.238: icmp_seq=3 ttl=118 time=14.400 ms
64 bytes from 142.250.182.238: icmp_seq=4 ttl=118 time=14.795 ms
64 bytes from 142.250.182.238: icmp_seq=5 ttl=118 time=51.621 ms
64 bytes from 142.250.182.238: icmp_seq=6 ttl=118 time=95.490 ms
64 bytes from 142.250.182.238: icmp_seq=7 ttl=118 time=14.350 ms
64 bytes from 142.250.182.238: icmp_seq=8 ttl=118 time=14.314 ms
64 bytes from 142.250.182.238: icmp_seq=9 ttl=118 time=16.518 ms
64 bytes from 142.250.182.238: icmp_seq=10 ttl=118 time=15.811 ms
64 bytes from 142.250.182.238: icmp_seq=11 ttl=118 time=18.380 ms
^C
--- google.com ping statistics ---
12 packets transmitted, 12 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 14.314/28.942/95.490/25.026 ms
(base) harsh@Harshs-MacBook-Air ~ %
```

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TCP.pcapng

icmp

No. | Time | Source | Destination | Protocol | Length | Info
---|---|---|---|---|---|---
7541 14.523078 142.250.183.238 172.16.73.21 ICMP 98 Echo (ping) request id=0xda64, seq=0/0, ttl=64 (reply in 7541)
7542 14.523078 172.16.73.21 142.250.183.238 ICMP 98 Echo (ping) reply id=0xda64, seq=0/0, ttl=115 (request in 7540)
7858 15.511122 172.16.73.21 142.250.183.238 ICMP 98 Echo (ping) request id=0xda64, seq=1/256, ttl=64 (reply in 7859)
7859 15.528851 142.250.183.238 172.16.73.21 ICMP 98 Echo (ping) reply id=0xda64, seq=1/256, ttl=115 (request in 7858)
8303 16.516319 172.16.73.21 142.250.183.238 ICMP 98 Echo (ping) request id=0xda64, seq=2/512, ttl=64 (reply in 8304)
8304 16.533688 142.250.183.238 172.16.73.21 ICMP 98 Echo (ping) reply id=0xda64, seq=2/512, ttl=115 (request in 8303)
8682 17.517800 172.16.73.21 142.250.183.238 ICMP 98 Echo (ping) request id=0xda64, seq=3/768, ttl=64 (reply in 8683)
8683 17.534434 142.250.183.238 172.16.73.21 ICMP 98 Echo (ping) reply id=0xda64, seq=3/768, ttl=115 (request in 8682)
9007 18.523089 172.16.73.21 142.250.183.238 ICMP 98 Echo (ping) request id=0xda64, seq=4/1024, ttl=64 (reply in 9007)
9054 18.563808 142.250.183.238 172.16.73.21 ICMP 98 Echo (ping) reply id=0xda64, seq=4/1024, ttl=115 (request in 9007)
9428 19.526934 172.16.73.21 142.250.183.238 ICMP 98 Echo (ping) request id=0xda64, seq=5/1280, ttl=64 (reply in 9428)
9429 19.543102 142.250.183.238 172.16.73.21 ICMP 98 Echo (ping) reply id=0xda64, seq=5/1280, ttl=115 (request in 9428)
9833 20.532090 172.16.73.21 142.250.183.238 ICMP 98 Echo (ping) request id=0xda64, seq=6/1536, ttl=64 (reply in 9834)
9834 20.578590 142.250.183.238 172.16.73.21 ICMP 98 Echo (ping) reply id=0xda64, seq=6/1536, ttl=115 (request in 9833)
10200 21.537285 172.16.73.21 142.250.183.238 ICMP 98 Echo (ping) request id=0xda64, seq=7/1792, ttl=64 (reply in 10224)
10224 21.621742 142.250.183.238 172.16.73.21 ICMP 98 Echo (ping) reply id=0xda64, seq=7/1792, ttl=115 (request in 10200)
10542 22.542478 172.16.73.21 142.250.183.238 ICMP 98 Echo (ping) request id=0xda64, seq=8/2048, ttl=64 (reply in 10556)
10556 22.560331 142.250.183.238 172.16.73.21 ICMP 98 Echo (ping) reply id=0xda64, seq=8/2048, ttl=115 (request in 10542)
10839 23.544450 172.16.73.21 142.250.183.238 ICMP 98 Echo (ping) request id=0xda64, seq=9/2304, ttl=64 (reply in 10886)
10886 23.567463 142.250.183.238 172.16.73.21 ICMP 98 Echo (ping) reply id=0xda64, seq=9/2304, ttl=115 (request in 10839)
11168 24.549166 172.16.73.21 142.250.183.238 ICMP 98 Echo (ping) request id=0xda64, seq=10/2560, ttl=64 (reply in 11169)

> Frame 7540: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface 0
> Ethernet II, Src: e2:8f:26:74:c2:ab (e2:8f:26:74:c2:ab), Dst: BrocadeCommu_90:80:78 (74:
> Internet Protocol Version 4, Src: 172.16.73.21, Dst: 142.250.183.238
> Internet Control Message Protocol

0000 74 8e f8 90 80 78 e2 8f 26 74 c2 ab 08 00 45 00 t...x...&t...E
0010 00 54 e8 4e 00 00 40 01 56 4c ac 10 49 15 8e fa .T.N.@.VL..I..
0020 b7 ee 08 00 c0 5b da 64 00 00 69 67 71 fb 00 04 .....[.d..igq
0030 96 d5 08 09 0a 0b 0c 0d 0e 0f 10 11 12 13 14 15 .....!#$%
0040 16 17 18 19 1a 1b 1c 1d 1e 1f 20 21 22 23 24 25 .....&'()*+,-./012345
0050 26 27 28 29 2a 2b 2c 2d 2e 2f 30 31 32 33 34 35 .....67
0060 36 37
```

2. To analyze TCP packets and observe the TCP three-way handshake

tcp									
No.	Time	Source	Destination	Protocol	Length	Info			
13328	30.107594	172.16.73.21	17.248.239.64	TCP	78	60714 → 443 [SYN, ECE, CWR] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=2542875075 TSecr=0 SA			
13371	30.136807	17.248.239.64	172.16.73.21	TCP	74	443 → 60714 [SYN, ACK, ECE] Seq=0 Ack=1 Win=14480 Len=0 MSS=1460 SACK_PERM TSval=712110116 T			
13373	30.137028	172.16.73.21	17.248.239.64	TCP	66	60714 → 443 [ACK] Seq=1 Ack=1 Win=131776 Len=0 TSval=2542875105 TSecr=712110116			
13374	30.138323	172.16.73.21	17.248.239.64	TCP	1514	60714 → 443 [ACK] Seq=1 Ack=1 Win=131776 Len=1448 TSval=2542875106 TSecr=712110116 [TCP PDU			
13375	30.138349	172.16.73.21	17.248.239.64	TLSSv1..	169	Client Hello (SNI=p176-contacts.icloud.com)			
13376	30.143633	17.248.239.64	172.16.73.21	TCP	66	443 → 60714 [ACK] Seq=1 Ack=1449 Win=17408 Len=0 TSval=712110119 TSecr=2542875106			
13377	30.143636	17.248.239.64	172.16.73.21	TCP	66	443 → 60714 [ACK] Seq=1 Ack=1552 Win=17408 Len=0 TSval=712110119 TSecr=2542875106			
13436	30.237296	17.248.239.64	172.16.73.21	TLSSv1..	1514	Server Hello, Change Cipher Spec, Application Data			
13437	30.237299	17.248.239.64	172.16.73.21	TCP	1514	443 → 60714 [PSH, ACK] Seq=1449 Ack=1552 Win=17408 Len=1448 TSval=712110126 TSecr=2542875106			
13438	30.237301	17.248.239.64	172.16.73.21	TCP	1266	443 → 60714 [PSH, ACK] Seq=2897 Ack=1552 Win=17408 Len=1200 TSval=712110126 TSecr=2542875106			
13439	30.237304	17.248.239.64	172.16.73.21	TLSSv1..	1155	Application Data, Application Data, Application Data			
13440	30.237482	172.16.73.21	17.248.239.64	TCP	66	60714 → 443 [ACK] Seq=1552 Ack=5186 Win=126592 Len=0 TSval=2542875205 TSecr=712110126			
13441	30.237655	172.16.73.21	17.248.239.64	TCP	66	[TCP Window Update] 60714 → 443 [ACK] Seq=1552 Ack=5186 Win=131072 Len=0 TSval=2542875206 TS			
13592	30.460892	172.16.73.21	17.248.239.64	TLSSv1..	146	Change Cipher Spec, Application Data			
13593	30.462718	172.16.73.21	17.248.239.64	TCP	1514	60714 → 443 [ACK] Seq=1632 Ack=5186 Win=131072 Len=1448 TSval=2542875431 TSecr=712110126 [TC			
13594	30.462721	172.16.73.21	17.248.239.64	TCP	1514	60714 → 443 [ACK] Seq=3080 Ack=5186 Win=131072 Len=1448 TSval=2542875431 TSecr=712110126 [TC			
13595	30.462745	172.16.73.21	17.248.239.64	TLSSv1..	1257	Application Data			
13596	30.512973	172.16.73.21	17.248.239.64	TCP	1514	[TCP Retransmission] 60714 → 443 [PSH, ACK] Seq=4271 Ack=5186 Win=131072 Len=1448 TSval=2542			
13647	30.566504	17.248.239.64	172.16.73.21	TCP	66	443 → 60714 [ACK] Seq=5186 Ack=1632 Win=17408 Len=0 TSval=712110159 TSecr=2542875429			
13648	30.568745	17.248.239.64	172.16.73.21	TCP	66	443 → 60714 [ACK] Seq=5186 Ack=3080 Win=20480 Len=0 TSval=712110162 TSecr=2542875431			
13649	30.568748	17.248.239.64	172.16.73.21	TCP	66	443 → 60714 [ACK] Seq=5186 Ack=4528 Win=23296 Len=0 TSval=712110162 TSecr=2542875431			

> Frame 13328: Packet, 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interf. 0000 74 8e f8 90 80 78 e2 f8 26 74 c2 ab 08 00 45 00 t...x...&t...E-
> Ethernet II, Src: e2:8f:26:74:c2:ab (e2:8f:26:74:c2:ab), Dst: BrocadeCommu_90:80:78 (74:8e:f8:90: 0010 00 40 00 00 40 00 40 06 44 5a ac 10 49 15 11 f8 @...@...DZ...I...
> Internet Protocol Version 4, Src: 172.16.73.21, Dst: 17.248.239.64 0020 ef 40 ed 2a 01 bb f1 1f f5 de 00 00 00 00 b0 c2 @*...*...
> Transmission Control Protocol, Src Port: 60714, Dst Port: 443, Seq: 0, Len: 0 0030 ff ff a0 a4 00 00 02 04 05 b4 01 03 03 06 01 01
0040 08 0a 97 91 31 c3 00 00 00 00 04 02 00 001.....

> Frame 13328: Packet, 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interface en0, 0000 74 8e f8 90 80 78 e2 f8 26 74 c2 ab 08 00 45 00 t...x...&t...E- > Ethernet II, Src: e2:8f:26:74:c2:ab (e2:8f:26:74:c2:ab), Dst: BrocadeCommu_90:80:78 (74:8e:f8:90: 0010 00 40 00 00 40 00 40 06 44 5a ac 10 49 15 11 f8 @...@...DZ...I... > Internet Protocol Version 4, Src: 172.16.73.21, Dst: 17.248.239.64 0020 ef 40 ed 2a 01 bb f1 1f f5 de 00 00 00 00 b0 c2 @*...*... > Transmission Control Protocol, Src Port: 60714, Dst Port: 443, Seq: 0, Len: 0 0030 ff ff a0 a4 00 00 02 04 05 b4 01 03 03 06 01 01 0040 08 0a 97 91 31 c3 00 00 00 00 04 02 00 001.....									
Source Port: 60714 Destination Port: 443 [Stream index: 0] [Stream Packet Number: 1] > [Conversation completeness: Complete, WITH_DATA (31)] [TCP Segment Len: 0] Sequence Number: 0 (relative sequence number) Sequence Number (raw): 4045403614 [Next Sequence Number: 1 (relative sequence number)] Acknowledgment Number: 0 Acknowledgment number (raw): 0 1011 = Header Length: 44 bytes (11) > Flags: 0x0c2 (SYN, ECE, CWR) Window: 65535 [Calculated window size: 65535] Checksum: 0xa0a4 [unverified] [Checksum Status: Unverified] Urgent Pointer: 0 > Options: (24 bytes), Maximum segment size, No-Operation (NOP), Window scale, No-Operation (NOP) > [Timestamps] [Client Contiguous Streams: 1] [Server Contiguous Streams: 1]									

3. To capture and analyze UDP packets

udp									
No.	Time	Source	Destination	Protocol	Length	Info			
8968	17.321967	172.16.41.135	172.31.1.6	DNS	91	Standard query 0x8464 AAAA messaging.engagement.office.com			
8969	17.323584	172.31.1.6	172.16.41.135	DNS	182	Standard query response 0x1d09 A messaging.engagement.office.com CNAME prod-campaignaggregator.omexexternal1fb.office.net.akad..			
8970	17.331977	173.194.154.38	172.16.41.135	UDP	1296	443 → 51085 Len=1250			
8972	17.337200	172.31.1.6	172.16.41.135	DNS	194	Standard query response 0x8464 AAAA messaging.engagement.office.com CNAME prod-campaignaggregator.omexexternal1fb.office.net.a..			
8974	17.344875	142.250.195.46	172.16.41.135	UDP	70	443 → 59855 Len=24			
8975	17.345072	172.16.41.135	142.250.195.46	UDP	73	59855 → 443 Len=31			
8978	17.345165	172.16.41.135	173.194.154.38	UDP	75	51085 → 443 Len=33			
8980	17.347519	fe80::aa1:89ff:fe55::ff02::fb	ff02::fb	MDNS	424	Standard query 0x0000 ANY HIKVISION DS-2CD3651G0-IZ - F60580171.._http..tcp.local, "QM" question ANY HIKVISION DS-2CD3651G0-IZ ..			
8982	17.347808	172.16.33.33	224.0.0.251	MDNS	400	Standard query 0x0000 ANY HIKVISION DS-2CD3651G0-IZ - F60580171.._http..tcp.local, "QM" question ANY HIKVISION DS-2CD3651G0-IZ ..			
8986	17.382648	172.16.33.82	224.0.0.251	MDNS	429	Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G38459834.._http..tcp.local, "QM" question ANY HIKVISION DS-2CD3143G0-IU ..			
8987	17.383073	fe80::2628:fdfff:fe02::fb	ff02::fb	MDNS	453	Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G38459834.._http..tcp.local, "QM" question ANY HIKVISION DS-2CD3143G0-IU ..			
8988	17.388697	142.250.195.46	172.16.41.135	UDP	114	443 → 59855 Len=68			
8989	17.397498	142.250.195.46	172.16.41.135	UDP	67	443 → 59855 Len=21			
8990	17.397831	172.16.41.135	142.250.195.46	UDP	77	59855 → 443 Len=35			
8991	17.412490	fe80::2628:fdfff:fe02::fb	ff02::fb	MDNS	424	Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G31130577.._http..tcp.local, "QM" question ANY HIKVISION DS-2CD3143G0-IU ..			
8992	17.412668	172.16.33.79	224.0.0.251	MDNS	400	Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G31130577.._http..tcp.local, "QM" question ANY HIKVISION DS-2CD3143G0-IU ..			
8993	17.413591	173.194.154.38	172.16.41.135	UDP	1296	443 → 51085 Len=1250			
8994	17.418481	172.16.36.131	224.0.0.251	MDNS	349	Standard query response 0x0000 PTR, cache flush iPhone-9.local PTR, cache flush iPhone-9.local PTR, cache flush iPhone-9.local..			
8995	17.419113	fe80::1805:a078:e19::ff02::fb	ff02::fb	MDNS	373	Standard query response 0x0000 PTR, cache flush iPhone-9.local PTR, cache flush iPhone-9.local PTR, cache flush iPhone-9.local..			
8996	17.419690	172.16.36.119	224.0.0.251	MDNS	1047	Standard query 0x0000 PTR_rdlink..tcp.local, "QM" question PTR_companion-link..tcp.local, "QM" question PTR_Akshin's iPad (2..			
8997	17.421680	fe80::1c52:6de9:a09::ff02::fb	ff02::fb	MDNS	1071	Standard query 0x0000 PTR_rdlink..tcp.local, "QM" question PTR_companion-link..tcp.local, "QM" question PTR_Akshin's iPad (2..			
8998	17.423066	172.16.33.86	224.0.0.251	MDNS	429	Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G38459843.._http..tcp.local, "QM" question ANY HIKVISION DS-2CD3143G0-IU ..			
8999	17.423962	fe80::2628:fdfff:fe02::fb	ff02::fb	MDNS	453	Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G38459843.._http..tcp.local, "QM" question ANY HIKVISION DS-2CD3143G0-IU ..			
9000	17.426562	fe80::2628:fdfff:fe02::fb	ff02::fb	MDNS	453	Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G38459850.._http..tcp.local, "QM" question ANY HIKVISION DS-2CD3143G0-IU ..			
9001	17.426789	172.16.33.80	224.0.0.251	MDNS	429	Standard query 0x0000 ANY HIKVISION DS-2CD3143G0-IU - G38459850.._http..tcp.local, "QM" question ANY HIKVISION DS-2CD3143G0-IU ..			
9002	17.428458	172.16.41.135	173.194.154.38	UDP	75	51085 → 443 Len=33			

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> Frame 8989: Packet, 67 bytes on wire (536 bits), 67 bytes captured (536 bits) on interface \Device\NPF_{30E8F03D-A225-4923-BF48-48BA90E0F267}, id 0
> Ethernet II, Src: Fortinet_6e:91:0f (04:d5:90:6e:91:0f), Dst: Intel_0a:fe:60 (98:43:fa:0a:fe:60)
  > Destination: Intel_0a:fe:60 (98:43:fa:0a:fe:60)
  > Source: Fortinet_6e:91:0f (04:d5:90:6e:91:0f)
    Type: 802.1Q Virtual LAN (0x8100)
    [Stream index: 4]
  > 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 0
    000. .... .... = Priority: Best Effort (default) (0)
    ...0 .... .... = DEI: Ineligible
    .... 0000 0000 0000 = ID: 0
    Type: IPv4 (0x0800)
  > Internet Protocol Version 4, Src: 142.250.195.46, Dst: 172.16.41.135
  > User Datagram Protocol, Src Port: 443, Dst Port: 59855
    Source Port: 443
    Destination Port: 59855
    Length: 29
    Checksum: 0xe883 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 368]
    [Stream Packet Number: 15]
    [Timestamps]
    UDP payload (21 bytes)
  > Data (21 bytes)

0000 98 43 fa 0a fe 60 04 d5 90 6e 91 0f 81 00 00 00  C...n....
0010 08 00 45 80 00 31 00 00 40 00 3c 11 16 7c 8e fa  ..E...@<...|..
0020 c3 2e ac 10 29 87 01 bb e9 cf 00 1d e8 83 50 1a  ....).....P.
0030 82 10 87 bc 1e e5 46 88 63 93 23 f6 2d 21 0a 13  ....F.c#-!...
0040 e9 d1 9c                                     ....

No.: 8989 - Time: 17.397498 - Source: 142.250.195.46 - Destination: 172.16.41.135 - Protocol: UDP - Length: 67 - Info: 443 -> 59855 Len=21

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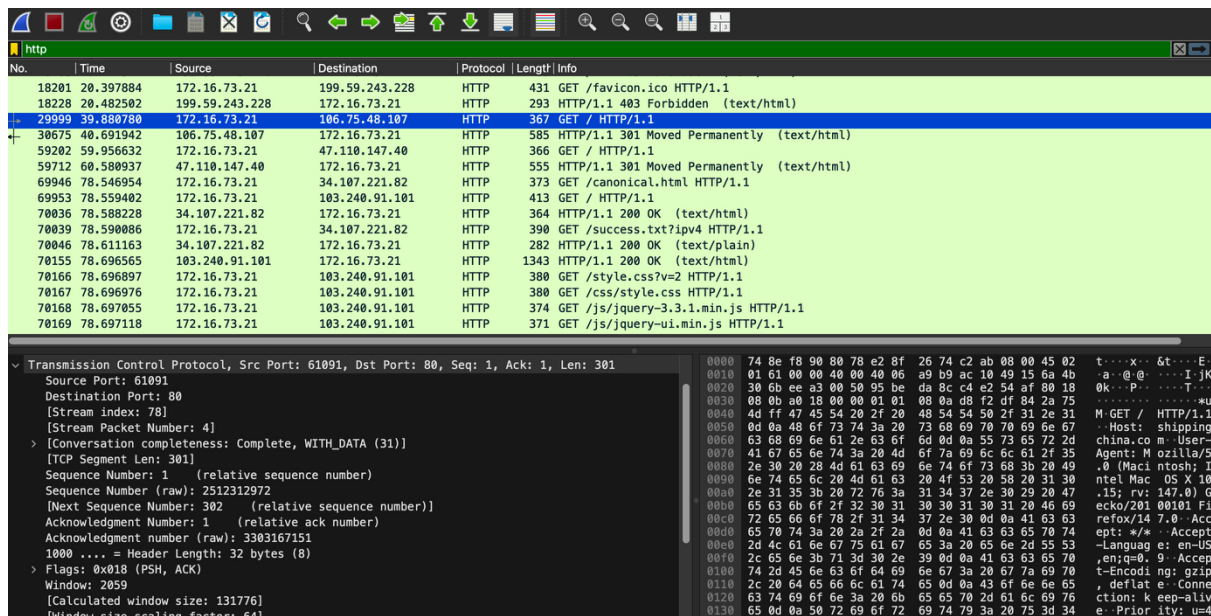
4. To capture and analyze DNS query and response packets

The screenshot displays the Wireshark network protocol analyzer. The top pane shows a list of captured packets, with the DNS protocol selected. The middle pane shows the details of the selected packet (No. 13372), including the Transaction ID (0xba5a) and the query type (Standard query). The bottom pane shows the raw packet data in hexadecimal and ASCII format.

No.	Time	Source	Destination	Protocol	Length	Info
7538	14.501498	172.16.73.21	172.31.1.6	DNS	70	Standard query 0x8676 A google.com
7539	14.509274	172.31.1.6	172.16.73.21	DNS	86	Standard query response 0x8676 A google.com A 142.250.183.238
7717	14.957882	172.16.73.21	172.31.1.6	DNS	100	Standard query 0xbc60 PTR db._dns-sd._udp.0.64.16.172.in-addr.arpa
7737	14.977299	172.31.1.6	172.16.73.21	DNS	100	Standard query response 0xbc60 No such name PTR db._dns-sd._udp.0.64.16.172.in-addr.arpa
8047	15.937440	172.16.73.21	172.31.1.6	DNS	99	Standard query 0xd3ed PTR b._dns-sd._udp.0.64.16.172.in-addr.arpa
8048	15.958402	172.31.1.6	172.16.73.21	DNS	99	Standard query response 0xd3ed No such name PTR b._dns-sd._udp.0.64.16.172.in-addr.arpa
9547	19.860251	172.16.73.21	172.31.1.6	DNS	79	Standard query 0x4c9a A time.cloudflare.com
9613	19.890664	172.31.1.6	172.16.73.21	DNS	111	Standard query response 0x4c9a A time.cloudflare.com A 162.159.200.1 A 162.159.200.123
13319	30.061537	172.16.73.21	172.31.1.6	DNS	84	Standard query 0x85ee HTTPS p176-contacts.icloud.com
13320	30.061677	172.16.73.21	172.31.1.6	DNS	84	Standard query 0x4cea A p176-contacts.icloud.com
13321	30.103529	172.31.1.6	172.16.73.21	DNS	206	Standard query response 0x85ee HTTPS p176-contacts.icloud.com CNAME contacts.fe2.apple-dns.net S
13322	30.103530	172.31.1.6	172.16.73.21	DNS	172	Standard query response 0x4cea A p176-contacts.icloud.com CNAME contacts.fe2.apple-dns.net A 17.
13323	30.104636	172.16.73.21	172.31.1.6	DNS	86	Standard query 0xba5a HTTPS contacts.fe2.apple-dns.net
13372	30.136809	172.31.1.6	172.16.73.21	DNS	171	Standard query response 0xba5a HTTPS contacts.fe2.apple-dns.net SOA ns-1396.awsdns-46.org
21281	47.729550	172.16.73.21	172.31.1.6	DNS	78	Standard query 0x3c0f HTTPS graph.whatsapp.net
21282	47.729864	172.16.73.21	172.31.1.6	DNS	78	Standard query 0xdf8e A graph.whatsapp.net
21350	47.845590	172.31.1.6	172.16.73.21	DNS	169	Standard query response 0x3c0f HTTPS graph.whatsapp.net CNAME static.whatsapp.net CNAME mmx-ds.c
21352	47.845595	172.31.1.6	172.16.73.21	DNS	140	Standard query response 0xdf8e A graph.whatsapp.net CNAME static.whatsapp.net CNAME mmx-ds.cdn.w
21354	47.847098	172.16.73.21	172.31.1.6	DNS	83	Standard query 0x332b HTTPS mmx-ds.cdn.whatsapp.net
21356	47.866293	172.31.1.6	172.16.73.21	DNS	128	Standard query response 0x332b HTTPS mmx-ds.cdn.whatsapp.net SOA a.ns.whatsapp.net

> Frame 13323: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface en0, ...
> Ethernet II, Src: e2:8f:26:74:c2:ab (e2:8f:26:74:c2:ab), Dst: BrocadeCommu_90:80:78 (74:8e:f8:90:78)
> Internet Protocol Version 4, Src: 172.16.73.21, Dst: 172.31.1.6
> User Datagram Protocol, Src Port: 50822, Dst Port: 53
> Domain Name System (query)
 Transaction ID: 0xba5a
 Flags: 0x0100 Standard query
 Questions: 1
 Answer RRs: 0
 Authority RRs: 0
 Additional RRs: 0
 Queries
 [Response In: 13372]

5. To capture and analyze HTTP packets



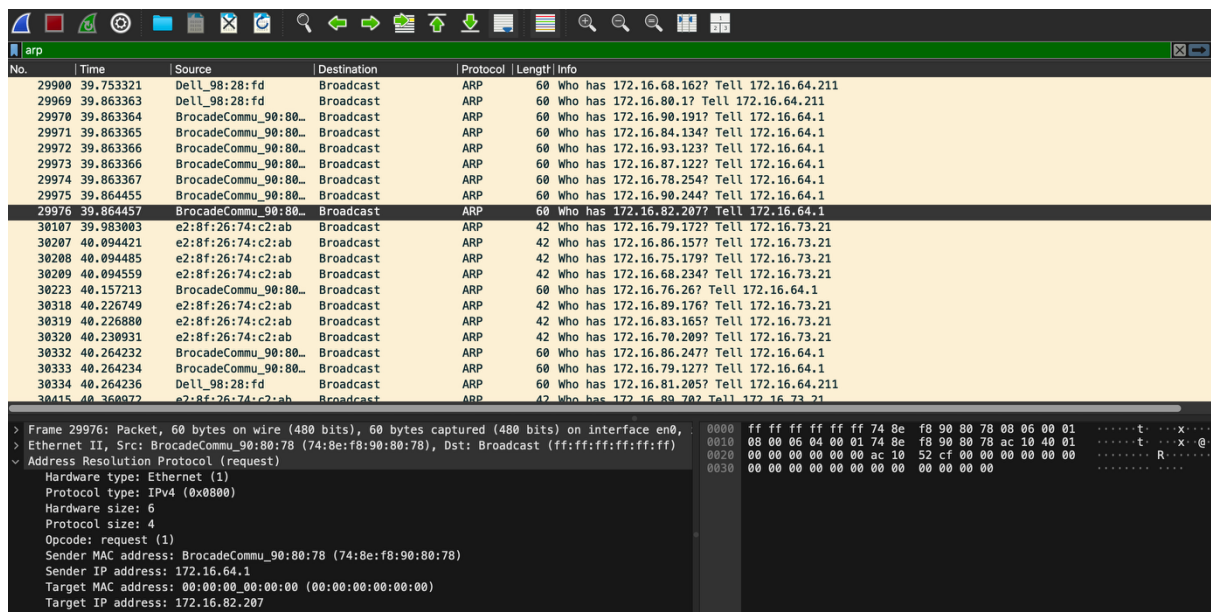
The image shows a Wireshark capture of HTTP traffic. The packet list on the left shows several HTTP GET requests and responses. The selected packet (No. 29999) is an HTTP GET request for / HTTP/1.1. The packet details pane on the left shows the structure of the HTTP request, including the method (GET), URI (/), and various headers. The packet bytes pane on the right shows the raw data of the packet, including the HTTP request line and headers.

No.	Time	Source	Destination	Protocol	Length	Info
18201	20.397884	172.16.73.21	199.59.243.228	HTTP	431	GET /favicon.ico HTTP/1.1
18228	20.482592	199.59.243.228	172.16.73.21	HTTP	293	HTTP/1.1 403 Forbidden (text/html)
29999	39.880780	172.16.73.21	106.75.48.107	HTTP	367	GET / HTTP/1.1
30675	40.691942	106.75.48.107	172.16.73.21	HTTP	585	HTTP/1.1 301 Moved Permanently (text/html)
59202	59.956632	172.16.73.21	47.110.147.40	HTTP	366	GET / HTTP/1.1
59712	60.580937	47.110.147.40	172.16.73.21	HTTP	555	HTTP/1.1 301 Moved Permanently (text/html)
69946	78.546954	172.16.73.21	34.107.221.82	HTTP	373	GET /canonical.html HTTP/1.1
69953	78.559402	172.16.73.21	103.240.91.101	HTTP	413	GET / HTTP/1.1
70036	78.588228	34.107.221.82	172.16.73.21	HTTP	364	HTTP/1.1 200 OK (text/html)
70039	78.590086	172.16.73.21	34.107.221.82	HTTP	390	GET /success.txt?ip=4 HTTP/1.1
70046	78.611163	34.107.221.82	172.16.73.21	HTTP	282	HTTP/1.1 200 OK (text/plain)
70155	78.696565	103.240.91.101	172.16.73.21	HTTP	1343	HTTP/1.1 200 OK (text/html)
70166	78.696897	172.16.73.21	103.240.91.101	HTTP	380	GET /style.css?v=2 HTTP/1.1
70167	78.696976	172.16.73.21	103.240.91.101	HTTP	380	GET /css/style.css HTTP/1.1
70168	78.697055	172.16.73.21	103.240.91.101	HTTP	374	GET /js/jquery-3.3.1.min.js HTTP/1.1
70169	78.697118	172.16.73.21	103.240.91.101	HTTP	371	GET /js/jquery-ui.min.js HTTP/1.1

Transmission Control Protocol, Src Port: 61091, Dst Port: 80, Seq: 1, Ack: 1, Len: 301

Source Port: 61091
Destination Port: 80
[Stream index: 78]
[Stream Packet Number: 4]
> [Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 301]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 2512312972
[Next Sequence Number: 302 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment Number (raw): 3303167151
1000 = Header Length: 32 bytes (8)
> Flags: 0x018 (PSH, ACK)
Window: 2059
[Calculated window size: 131776]
[Window size scaling factor: 64]

6. To analyze ARP request and reply packets



The image shows a Wireshark capture of ARP traffic. The packet list on the left shows several ARP requests and replies. The selected packet (No. 29976) is an ARP request. The packet details pane on the left shows the structure of the ARP request, including the hardware type, protocol type, hardware size, protocol size, opcode, sender MAC address, sender IP address, target MAC address, and target IP address. The packet bytes pane on the right shows the raw data of the packet, including the ARP request line and headers.

No.	Time	Source	Destination	Protocol	Length	Info
29900	39.753321	Dell_98:28:fd	Broadcast	ARP	60	Who has 172.16.68.162? Tell 172.16.64.211
29969	39.863363	Dell_98:28:fd	Broadcast	ARP	60	Who has 172.16.80.1? Tell 172.16.64.211
29970	39.863364	BrocadeComm_90:80:78	Broadcast	ARP	60	Who has 172.16.90.191? Tell 172.16.64.1
29971	39.863365	BrocadeComm_90:80:78	Broadcast	ARP	60	Who has 172.16.84.134? Tell 172.16.64.1
29972	39.863366	BrocadeComm_90:80:78	Broadcast	ARP	60	Who has 172.16.93.123? Tell 172.16.64.1
29973	39.863366	BrocadeComm_90:80:78	Broadcast	ARP	60	Who has 172.16.87.122? Tell 172.16.64.1
29974	39.863367	BrocadeComm_90:80:78	Broadcast	ARP	60	Who has 172.16.78.254? Tell 172.16.64.1
29975	39.864455	BrocadeComm_90:80:78	Broadcast	ARP	60	Who has 172.16.90.244? Tell 172.16.64.1
29976	39.864457	BrocadeComm_90:80:78	Broadcast	ARP	60	Who has 172.16.82.207? Tell 172.16.64.1
30107	39.983083	e2:8f:26:74:c2:ab	Broadcast	ARP	42	Who has 172.16.79.172? Tell 172.16.73.21
30207	40.094421	e2:8f:26:74:c2:ab	Broadcast	ARP	42	Who has 172.16.86.157? Tell 172.16.73.21
30208	40.094485	e2:8f:26:74:c2:ab	Broadcast	ARP	42	Who has 172.16.75.179? Tell 172.16.73.21
30209	40.094559	e2:8f:26:74:c2:ab	Broadcast	ARP	42	Who has 172.16.68.234? Tell 172.16.73.21
30223	40.157213	BrocadeComm_90:80:78	Broadcast	ARP	60	Who has 172.16.76.26? Tell 172.16.64.1
30318	40.226749	e2:8f:26:74:c2:ab	Broadcast	ARP	42	Who has 172.16.89.176? Tell 172.16.73.21
30319	40.226800	e2:8f:26:74:c2:ab	Broadcast	ARP	42	Who has 172.16.83.165? Tell 172.16.73.21
30320	40.230931	e2:8f:26:74:c2:ab	Broadcast	ARP	42	Who has 172.16.70.209? Tell 172.16.73.21
30332	40.264232	BrocadeComm_90:80:78	Broadcast	ARP	60	Who has 172.16.86.247? Tell 172.16.64.1
30333	40.264234	BrocadeComm_90:80:78	Broadcast	ARP	60	Who has 172.16.79.127? Tell 172.16.64.1
30334	40.264236	Dell_98:28:fd	Broadcast	ARP	60	Who has 172.16.81.205? Tell 172.16.64.211
30415	40.360972	e2:8f:26:74:c2:ab	Broadcast	ARP	42	Who has 172.16.80.70? Tell 172.16.73.21

> Frame 29976: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface en0
> Ethernet II, Src: BrocadeComm_90:80:78 (74:8e:f8:90:80:78), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
> Address Resolution Protocol (request)
Hardware type: Ethernet (1)
Protocol type: IPv4 (0x0800)
Hardware size: 6
Protocol size: 4
Opcode: request (1)
Sender MAC address: BrocadeComm_90:80:78 (74:8e:f8:90:80:78)
Sender IP address: 172.16.64.1
Target MAC address: 00:00:00:00:00:00 (00:00:00:00:00:00)
Target IP address: 172.16.82.207

7. To identify source and destination MAC and IP addresses in captured packets.

The image shows a Wireshark packet capture analysis. The top pane displays a list of captured packets. The selected packet (No. 6892) is expanded, showing the following details:

- Ethernet II**: Src: Intel_0a:fe:60 (98:43:fa:0a:fe:60), Dst: Fortinet_6e:91:0f (04:d5:90:6e:91:0f)
 - Destination: Fortinet_6e:91:0f (04:d5:90:6e:91:0f)
 - Source: Intel_0a:fe:60 (98:43:fa:0a:fe:60)
 - Type: IPv4 (0x0800)
 - [Stream index: 4]
- Internet Protocol Version 4**: Src: 172.16.41.135, Dst: 13.204.106.34
 - Version: 4
 - Header Length: 20 bytes (5)
 - Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 - Total Length: 41
 - Identification: 0xba38 (47672)
 - Flags: 0x2, Don't fragment
 - Fragment Offset: 0
 - Time to Live: 128
 - Protocol: TCP (6)
 - Header Checksum: 0x0000 [validation disabled]
 - [Header checksum status: Unverified]
 - Source Address: 172.16.41.135
 - Destination Address: 13.204.106.34
 - [Stream index: 213]
- Transmission Control Protocol**: Src Port: 60059, Dst Port: 443, Seq: 1, Ack: 1, Len: 1

The bottom pane shows the raw packet data in hexadecimal and ASCII format.